

DNP3 Protocol Technical Reference Guide

1. Introduction to DNP3 & Operational Architecture

Distributed Network Protocol 3 (DNP3) is a robust, open-source communications protocol widely deployed in SCADA (Supervisory Control and Data Acquisition) systems, particularly in electric, water, oil, gas, and transportation utilities. Standardized under IEEE Std 1815, DNP3 was explicitly engineered for reliable operational communication over bandwidth-constrained, noisy, or unreliable physical media.

1.1 Key Architectural Principles

- **Master/Outstation Relationship:** Communication occurs between a central control system (**Master**) and field devices (**Outstations** such as RTUs, PLCs, or IEDs).
- **Time-Stamped Data Events:** Outstations record and time-stamp state changes or analog threshold crossings at the exact instant they occur (down to millisecond resolution), ensuring precise sequence-of-events (SOE) recording regardless of polling latency.
- **Report-by-Exception (RBE):** Outstations store state changes as *Events*. The Master only retrieves newly changed data rather than polling the entire memory table every cycle, dramatically reducing bandwidth usage.
- **Unsolicited Responses:** Unlike traditional master-slave protocols (like Modbus), a DNP3 Outstation can spontaneously initiate communication to report critical alarms or events without waiting for a Master request.
- **Data Class Categorization:** Data is segregated into four classes:
 - **Class 0:** Current static/instantaneous values for all configured points.
 - **Class 1:** High-priority event data (e.g., circuit breaker trips, critical safety alarms).
 - **Class 2:** Medium-priority event data (e.g., analog limit warnings).
 - **Class 3:** Low-priority event data (e.g., routine status updates, minor telemetry changes).

1.2 Enhanced Performance Architecture (EPA) Model

DNP3 uses a streamlined 3-layer architecture derived from the 7-layer OSI model, known as the **Enhanced Performance Architecture (EPA)**, with an auxiliary **Transport Function** layer.



2. Detailed Data Frame Formats

Data encapsulation in DNP3 flows top-down during transmission and bottom-up during reception. Each layer appends its specific header and integrity fields.

2.1 Data Link Layer Frame Structure

Every DNP3 Data Link frame has a maximum total size of **292 bytes**. To guarantee data integrity over noisy channels, the header and data payloads are protected by mandatory 16-bit CRC checks.

[illegible]

Field Name	Size	Description
Sync Bytes	2 Bytes	Fixed pattern <code>0x05 0x64</code> (00000101 01100100) indicating start of frame.
Length	1 Byte	Number of user octets following, including Control byte, Dest/Src addresses (Range: 5 to 255). Excludes CRC bytes.
Control Byte	1 Byte	Contains DIR (Direction), PRM (Primary), FCB (Frame Count Bit), FCV (Frame Count Valid), and Function Codes (ACK, NACK, Link Status).
Destination Address	2 Bytes	Target device DNP3 address (Little-Endian, 0x0000 to 0xFFFE; 0xFFFF = Broadcast).
Source Address	2 Bytes	Originating device DNP3 address (Little-Endian).
Header CRC	2 Bytes	16-bit CRC calculated over bytes 0 through 7 of the header using polynomial $x^{16} + x^{13} + x^{12} + x^{11} + x^{10} + x^8 + x^4 + x^2 + x + 1$.
Data Chunks & CRCs	Variable	User data divided into blocks of up to 16 bytes, each followed by its own 2-byte CRC. Maximum 250 data bytes per frame.

2.2 Transport Function Header Structure

Because Application Layer fragments can be up to **2048 bytes**, the Transport Function breaks them down into 250-byte chunks for the Data Link layer. It prepends a **1-byte Transport Header** to each chunk:

```

Bit Position: 7 6 5 4 3 2 1 0 +-----+-----+-----+-----+ | FIN |
FIR | Sequence Number | +-----+-----+-----+-----+

```

- **FIN (Final Bit):** Set to 1 on the last segment of an Application Fragment.
- **FIR (First Bit):** Set to 1 on the first segment of an Application Fragment.
- **Sequence Number (6 Bits):** Increments modulo-64 (0 to 63) to detect missing or duplicate frames.

2.3 Application Layer Message Formats

The Application Layer carries SCADA commands and measurements structured as Application Protocol Data Units (APDUs).

A. Application Request APDU (Master → Outstation)

```
+-----+-----+-----+-----+
App Control (1B) | Function Code(1B)| Object Header(s) (Variable) | Data Fields (Optional)|
+-----+-----+-----+-----+
```

B. Application Response APDU (Outstation → Master)

```
+-----+-----+-----+-----+
+-----+ | App Control (1B) | Function Code(1B)| Internal Ind. (2B) | Object
```

Internal Indications (IIN) Bitmask Breakdown (2 Bytes)

Every response from an Outstation includes a 16-bit IIN field indicating system status and diagnostic alarms:

- **IIN1.0:** Device Broadcast Received
- **IIN1.1:** Class 1 Data Available
- **IIN1.2:** Class 2 Data Available
- **IIN1.3:** Class 3 Data Available
- **IIN1.4:** Time Sync Required from Master
- **IIN1.5:** Outstation Outputs in Local Mode
- **IIN1.6:** Device Trouble / Hardware Fault
- **IIN1.7:** Outstation Restarted (Device Reset)
- **IIN2.0:** Function Code Not Supported
- **IIN2.1:** Object Unknown
- **IIN2.2:** Parameter Error / Out of Range
- **IIN2.3:** Buffer Overflow
- **IIN2.4:** Operation Already Executing
- **IIN2.5:** Configuration Corrupt

3. Function Codes, Object Groups, and Commands

3.1 Key Application Function Codes

Code (Hex)	Code Name	Direction	Description
0x01	READ	Master → Outstation	Request object data or class events.
0x02	WRITE	Master → Outstation	Write values (e.g., date/time, configuration).
0x03	SELECT	Master → Outstation	Arm a control output (Select-Before-Operate step 1).
0x04	OPERATE	Master → Outstation	Execute an armed control output (Select-Before-Operate step 2).
0x05	DIRECT_OPERATE	Master → Outstation	Execute control output immediately without Select step.
0x06	DIRECT_OPERATE_NO_ACK	Master → Outstation	Execute control output without returning response.
0x07	FREEZE	Master → Outstation	Copy current counter values to frozen memory.

Code (Hex)	Code Name	Direction	Description
0x0D	COLD_RESTART	Master → Outstation	Reboot the Outstation hardware completely.
0x14	ENABLE_UNSOLICITED	Master → Outstation	Enable unsolicited response messages for Class 1, 2, or 3.
0x81	RESPONSE	Outstation → Master	Standard response containing requested data or status.
0x82	UNSOLICITED_RESPONSE	Outstation → Master	Spontaneous message carrying event notifications.

3.2 Common DNP3 Object Groups & Variations

Group	Variation	Object Name	Description
Group 1	Var 1 / 2	Binary Input	Single-bit digital inputs (Var 1 = Without status flag, Var 2 = With status flag).
Group 2	Var 1 / 2	Binary Input Event	State change events (Var 1 = Without time, Var 2 = With absolute 48-bit time stamp).
Group 10	Var 1 / 2	Binary Output	Digital output status points.
Group 12	Var 1	Control Relay Output Block (CROB)	Control command object used for pulse/latch output control.
Group 30	Var 1 / 2 / 5 / 6	Analog Input	Measurements (32-bit int, 16-bit int, 32-bit float, 64-bit float).
Group 32	Var 1..8	Analog Input Event	Analog value change events with or without timestamps.
Group 50	Var 1	Time and Date	System clock representation (48-bit millisecond counter since UNIX epoch).
Group 60	Var 1 / 2 / 3 / 4	Class Data Objects	Class 0 (Var 1), Class 1 (Var 2), Class 2 (Var 3), Class 3 (Var 4).

4. DNP3 Secure Authentication (DNP3-SA / IEEE 1815 Specification)

Standard DNP3 lacks inherent authentication or cryptographic mechanisms, leaving legacy networks susceptible to unauthorized command injection, eavesdropping, and replay attacks. **DNP3 Secure Authentication (DNP3-SA)** introduces application-layer challenge-response architecture, session management, and cryptographic message authentication without altering the fundamental underlying data structures.

4.1 Core Security Objectives

- **Authenticity:** Confirms the identity of the Master or Outstation making the request before executing critical actions.
- **Integrity Verification:** Ensures that control payloads and operational data have not been modified or corrupted in transit.

- **Replay Attack Prevention:** Uses cryptographically generated random challenges (nonces) and rolling sequence counters to block replay of captured valid frames.
- **Low Computational Overhead:** Avoids encrypting non-sensitive routine telemetry (e.g., periodic voltage reads), applying cryptographic hashing primarily to control operations and security-critical configuration events.

4.2 Key Hierarchy & Cryptographic Algorithms

DNP3-SA employs a two-tier key management structure:

Key Type	Lifetime	Function & Management
Update Keys (User/Master)	Long-Term (Months/ Years)	Symmetric pre-shared keys or asymmetric public/private keys stored securely in non-volatile memory. Used solely to encrypt and distribute temporary Session Keys.
Session Keys	Short-Term (Hours/ Days or N messages)	Symmetric keys dynamically generated and updated using Session Key Status / Change APDUs. Divided into <i>Control Directions</i> (Master-to-Outstation MAC key vs Outstation-to-Master MAC key).

Supported Cryptographic Algorithms (IEEE 1815 SBO/SA Suites)

- **HMAC-SHA-1-48:** Legacy truncated HMAC mode (6 bytes MAC).
- **HMAC-SHA-256-128:** Standard industrial baseline (16 bytes MAC output derived from SHA-256).
- **AES-GMAC-128 / AES-GMAC-256:** Galois Message Authentication Code mode for ultra-fast hardware-accelerated processing.

4.3 Challenge-Response Authentication Mechanism

When a Master issues a critical command (e.g., CROB Direct Operate), DNP3-SA triggers an application-layer Challenge-Response flow:

```

MASTER OUTSTATION | | | ----- 1. Critical Request (e.g., Operate Relay #2) -----> |
(Intercepts critical command) | | | <----- 2. AUTHENTICATION_CHALLENGE (Func 0x83 / Obj 120 Var 1)
- | (Contains Random Nonce & Seq #) | | | (Calculates HMAC over Command + Nonce using Session
Key) | | | ----- 3. AUTHENTICATION_REPLY (Func 0x1F / Obj 120 Var 2) -----> | (Includes HMAC
MAC Digest) | | | (Verifies HMAC calculation) | <----- 4. SUCCESS RESPONSE / EXECUTION
CONFIRMATION -----> | (Executes command & replies) | |

```

4.4 Security Application Objects (Group 120)

Object Group	Variation	Name	Description
Group 120	Var 1	Authentication Challenge	Contains Challenge Sequence Number, User Number, MAC Algorithm ID, and Random Nonce Data.
Group 120	Var 2	Authentication Reply	Contains Challenge Sequence Number, User Number, and the calculated Message Authentication Code (MAC).
Group 120	Var 3	Aggressive Mode Request	Includes pre-computed MAC directly inside the request to execute control in a single round-trip latency.
Group 120	Var 4	Session Key Status Request	Requests current validity status and challenge data for updating short-term session keys.

Object Group	Variation	Name	Description
Group 120	Var 5	Session Key Status	Outstation response containing Key Change Sequence Number and key status details.
Group 120	Var 6	Key Change Request	Transmits newly encrypted Session Keys from Master to Outstation using Update Key wrapping.
Group 120	Var 7	Authentication Error	Reports authentication failures, invalid MACs, sequence mismatches, or key expiration events.

5. Physical Communication Mediums: Serial vs Ethernet TCP/IP

DNP3 operates seamlessly over both legacy serial communication buses and modern IP networks.

Parameter	Serial Media (RS-232 / RS-485)	Ethernet Media (TCP/IP or UDP/IP)
Physical Layer	EIA/TIA-232, RS-485 2-Wire or 4-Wire, Fiber Optic	100Base-TX / 1000Base-T Ethernet, Cellular, Microwave
Default Port	Physical COM Port / DB9 / Terminal Block	TCP/UDP Port 20000 (IANA Standard)
Framing Handling	Raw octet stream, rely on 0x05 0x64 framing and CRCs	DNP3 frame encapsulated inside standard TCP/IP packet
Duplex Mode	RS-485 is Half-Duplex; RS-232 is Full-Duplex	Full-Duplex
Topology	Point-to-Point or Multi-drop Bus (RS-485)	Star, Ring, Mesh IP networks

6. Detailed Step-by-Step Configuration Guide

6.1 Serial Connection Configuration Steps (RS-485 / RS-232)

Step 1: Physical Wiring & Electrical Specification

- **RS-485 2-Wire Setup:** Connect Data+ (A) to Data+ (A), Data- (B) to Data- (B), and Signal Ground (SG) across all devices. Attach 120Ω termination resistors at the two extreme physical ends of the bus.
- **RS-232 Setup:** Connect TXD to RXD, RXD to TXD, and GND to GND (Null Modem setup).

Step 2: Serial UART Parameter Matching

Configure the serial interface parameters identically on both Master and Outstation:

- **Baud Rate:** Standard rates (e.g., 9600, 19200, 38400, 115200 bps). Default industrial recommendation: **9600 or 19200 bps** for long serial lines.
- **Data Bits:** Exactly **8 Bits** (Mandatory for binary DNP3 protocol).
- **Parity:** **None** or **Even** (Must match across all nodes).
- **Stop Bits:** **1 Stop Bit**.
- **Flow Control:** None, or Hardware (RTS/CTS) if using radio modems.

Step 3: DNP3 Link Layer & Timing Parameters

- **DNP3 Addresses:** Set Master Address (e.g., 1) and Outstation Address (e.g., 2). Each Outstation on a multi-drop RS-485 bus MUST have a unique address (0 to 65534).
- **Link Confirmations:** Enable or Disable Data Link layer ACKs. (Disabled by default over reliable links to minimize traffic).
- **Line Turnaround Delay:** Set RTS-to-TX delay (e.g., 10–50 ms) to allow RS-485 transceivers to switch from receive to transmit mode.

6.2 Ethernet TCP/IP Connection Configuration Steps

Step 1: Network IP Addressing & Sockets

- Assign static IP addresses, Subnet Mask, and Gateway to both Master station and Outstation hardware (e.g., Outstation IP: 192.168.1.50 , Master IP: 192.168.1.10).
- Set target TCP/UDP Port to **20000** on the Outstation (listening server socket).

Step 2: TCP Architecture Selection (Client vs. Server Roles)

- **Master Initiated (Standard):** Outstation acts as *TCP Server* (listening on port 20000). Master acts as *TCP Client* and initiates connection.
- **Dual-Endpoint / Dual-Socket (Dual-Initiated):** Used when Outstation needs to send Unsolicited Responses through firewalls. Both Master and Outstation can open TCP connections.

Step 3: TCP Keep-Alive & Socket Timeouts

- **TCP Keep-Alive:** Set interval (e.g., 30–60 seconds) to detect dropped cellular or IP connections without sending DNP3 polls.
- **Connect Timeout:** Set TCP connection establishment timeout (e.g., 5000 ms).
- **Application Layer Confirm Timeout:** Set time to wait for DNP3 response (e.g., 3000 ms) before retrying or declaring device offline.

Security Recommendation

Standard DNP3 traffic over TCP/IP is unencrypted and vulnerable to spoofing or MITM attacks. For production networks, implement **DNP3 Secure Authentication (DNP3-SA / IEEE 1815a)** or wrap port 20000 traffic inside an **IPsec / TLS VPN tunnel**.